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Review



Why some hearts heal and others don't: The phylogenetic landscape of cardiac regenerative capacity

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ABSTRACT

The limited ability of adult humans to replenish lost heart muscle cells after a heart attack has attracted scientists to explore natural heart regeneration capabilities in the animal kingdom. In particular, research has accelerated since the landmark discovery more than twenty years ago that zebrafish can completely regrow myocardial tissue. In this review, we survey heart regeneration studies in diverse model and non-model animals, aiming to gain insights into both the evolutionary trends in cardiac regenerative potential and the variations among closely related species. Differences in cardiomyogenesis, vasculature formation, and the communication between cardiovascular cells and other players have been investigated to understand the cellular basis, although the precise molecular and genetic causes underlying the stark differences in cardiac regenerative potential among certain close cousins remain largely unknown. By studying cardiovascular regeneration and repair in diverse organisms, we may uncover distinct mechanisms, offering new perspectives for advancing regenerative medicine.

1. Introduction

Cardiovascular diseases are among the leading causes of death worldwide, accounting for approximately one-third of all deaths [1]. One of the causes of cardiac disease is ischemic injury following myocardial infarction. Although medical advancements have improved patient survival, a fundamental cure is still needed, especially for chronic ischemic heart diseases, which remain irreversible with current treatments [2,3]. This challenge is attributed to the fact that cardiomyocytes are terminally differentiated and cannot proliferate [4]. Recent clinical approaches, such as stem cell therapy and pharmacological treatments, aim to promote proliferation, but most face challenges [5,6]. To establish these therapies, it is important to understand the mechanisms, particularly the target factors and their related molecular pathways.

Certain species, such as teleost fish and amphibians, can regenerate their hearts through the proliferation of diploid cardiomyocytes, highlighting a stark contrast to mammals like mice and humans [7,8]. Although traditional models like mice and zebrafish have provided significant insights into heart regeneration, the mechanisms discovered in these species are difficult to translate to humans, possibly due to side effects and incomplete understanding. This has driven researchers to seek novel insights from non-model organisms that may possess unexplored innate regenerative mechanisms. These animals, while lacking genomic tools, may have unique regenerative abilities that provide new perspectives. This review explores the potential of these non-model organisms (Fig. 1) and how their cardiomyocyte and vascular states could reveal new avenues for heart regeneration. By studying these species, we may uncover new strategies for treating heart disease in humans.

2. Animal models of cardiovascular regeneration studies

2.1. Fish

2.1.1. Zebrafish vs medaka fish

Zebrafish (*Danio rerio*), renowned as one of the most well-studied regenerative vertebrates, exhibits a remarkable ability to regenerate its heart after apical resection. This regeneration occurs primarily through the activation of cardiomyocyte proliferation, allowing the heart to heal effectively following injury [9]. Interestingly, the medaka (*Oryzias latipes*), despite being closely related, does not share this regenerative capability. In medaka, when cardiomyocytes are lost, the missing regions are replaced by fibroblast cells, which do not contribute to functional heart muscle [10]. However, medaka can regenerate its fin

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efficiently [11], suggesting that its loss of cardiac regenerative potential is tissue-specific.

2.1.2. *Mexican cavefish and surface fish*

Organisms evolve to adapt to their environments, even when they are genetically close. A fascinating example is the fish species *Astyanax mexicanus*, which has two distinct populations: one that lives in rivers and another that resides in caves. Over the course of approximately 3 million years, the cave-dwelling fish have lost several notable features, including their eyes, pigmentation, and, intriguingly, their heart regenerative capacity [13,14]. While the surface-dwelling fish can regenerate their hearts within a few months—similar to the regenerative abilities observed in zebrafish—the cavefish exhibit no such capability.

2.2. *Amphibian*

2.2.1. *Xenopus tropicalis vs Xenopus laevis*

Amphibians, such as newts and axolotls, are renowned for their remarkable regenerative capacity, particularly as tetraploid species [15,16]. In addition to these well-known animals, *Xenopus* frogs have also gained popularity in regeneration studies due to their unique biology and regenerative abilities. Two species of *Xenopus* commonly utilized in research are *Xenopus laevis* and *Xenopus tropicalis*. The primary distinction between these two species lies in their genome ploidy; *X. laevis* is pseudo-tetraploid, while *X. tropicalis* is diploid, similar to humans [17]. This difference in ploidy makes them intriguing comparative models for a variety of studies, including tissue regeneration. Both species of *Xenopus* exhibit impressive regenerative abilities, especially during their tadpole stage, where they can regenerate structures such as tails and limbs [18]. However, their capabilities diverge significantly when it comes to heart regeneration. Following apical resection of the heart, *X. tropicalis* is able to regenerate its heart tissue completely without leaving any scar tissue [19]. In contrast, *X. laevis* displays considerable fibrosis and is unable to regenerate its heart effectively [20]. This divergence in regenerative capacity is evident at the cellular level as well. *X. tropicalis* demonstrates a marked increase in cardiomyocyte proliferation at the injured border zone after apex resection, which is crucial for facilitating heart regeneration. Conversely, *X. laevis* develops a hypertrophic phenotype in response to injury but does not show an increase in the expression of proliferation-related genes. These findings strongly suggest that cardiomyocyte proliferation following apex resection is a critical event for heart regeneration in frogs. Despite these insights, there remains much to explore regarding the differences between these two species. One caveat of this comparison is that studies of cardiac injury responses in these two species were performed in two laboratories. Differences in the age of animals and clot size, rather than species differences, have been discussed as possible key contributors to the different outcomes [21]. Thus, it is essential to compare the cardiac regenerative potential of these two species by rigorous side-by-side experiments.

2.3. *Reptile*

It is well-known that lizards, such as geckos, possess the remarkable ability to autotomize their tails as a defense mechanism, allowing them to escape from predators [22]. This fascinating trait raises the hypothesis that geckos may also have the capacity to regenerate other tissues and organs. Recent research has provided compelling evidence of the impressive proliferation capacity of the gecko *Eublepharis macularius* [23,24]. In this species, cardiomyocytes are observed to proliferate continuously during the juvenile stage, highlighting a potential for heart regeneration. Indeed, adult gecko cardiomyocytes have been reported to mount robust regenerative responses following a puncture wound injury [25]. Whether heart regeneration is complete and scar-free in adult geckos awaits further evidence.

2.4. *Bird*

There are only a handful of reports on cardiac regeneration and repair in birds. A study focused on *Gallus domesticus* (domestic chickens) revealed that 3- to 5-day-old embryos exhibit remarkable regenerative capacity in their hearts following myocardial resection [26,27]. This impressive regenerative ability, however, is significantly reduced or even absent in 18-day-old embryos and newly hatched chicks, where scar formation becomes evident [27,28]. Adult pigeons also show limited myocardial regeneration. While they show some restoration of injured myocardium through the myofibrillar apparatus, the newly formed myofibrils tend to collapse over time, indicating a more limited capacity for regeneration compared to embryonic stages [29]. These observations suggest that the regenerative capacity of birds appears to

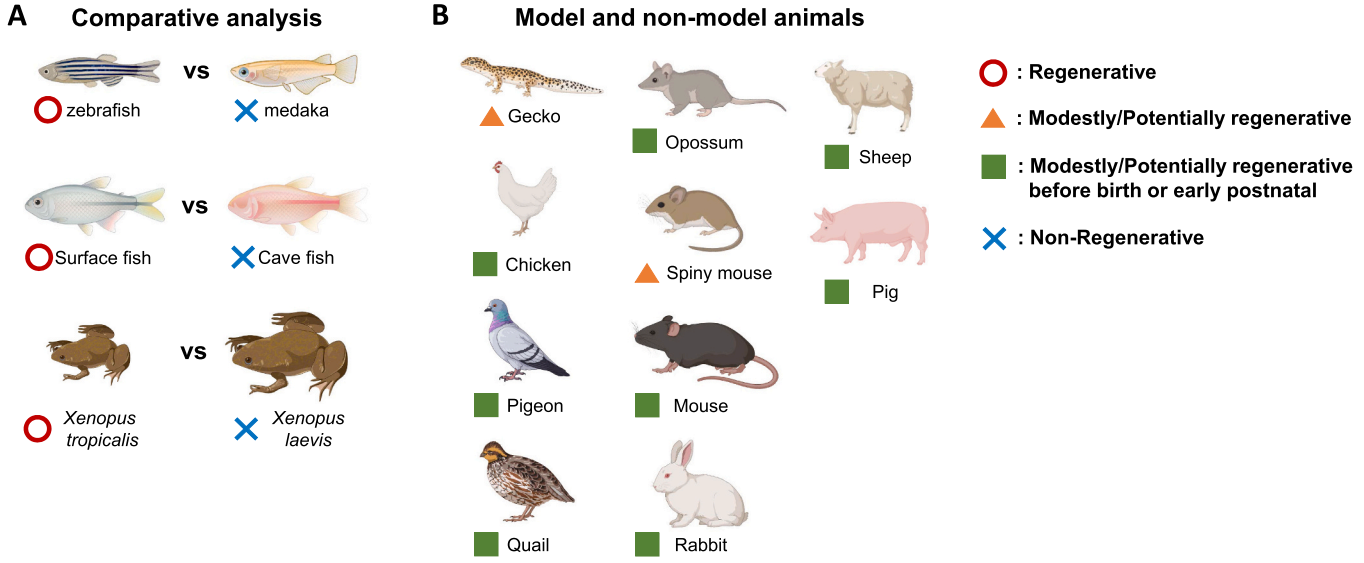


Fig. 1. Cardiac regenerative capacity varies considerably across species. (A) Despite similarities in body size or physiological characteristics, there are significant differences in the regenerative potential of the heart across species. (B) Both model and non-model animals have been investigated to assess their regenerative abilities, focusing on the proliferative capacity of cardiomyocytes and vascular cells, among other cellular characteristics. However, the underlying mechanisms driving these differences remain largely unexplored.

be restricted to a narrow window before the hatching period, after which it declines markedly. However, inducing proper heart injury in newborn and adult birds requires damages of their rigid, plate-like ribcage to access the heart, which may impede heart regeneration. Nevertheless, understanding the dynamic bird regenerative potential in different stages could provide valuable insights into the evolutionary biology of regeneration.

2.5. Mammal

2.5.1. Opossum

Mammals are broadly classified into three major groups: monotremes, marsupials, and placental mammals [30]. Among these, the opossum is considered a relatively ancestral species within the marsupial category, and its genome has recently been published, providing valuable insights into its biology [31]. Notably, the grey short-tailed opossum (*Monodelphis domestica*) has been reported to exhibit cardiac regenerative ability at postnatal day 14 (P14), whereas this capacity is lost by P29 [32]. This postnatal regenerative window is much longer than any other mammal reported to date (see below), which may be attributed to their altricial nature - newborn opossums resemble embryonic mammals more closely.

2.5.2. Mouse

Mice lose their capacity for heart regeneration much earlier postnatally [4]. Seminal studies by Porrelo et al. reported that newborn mice (P1) can fully regrow the lost or damaged myocardium after either a ventricle apex resection or myocardial infarction injury [4,33]. Heart function was restored and almost no scar was observed [4,33]. However, such a regenerative capacity was completely lost after P7 [4,33], which is consistent with the general consensus that the adult mouse heart has limited cardiac regenerative capacity.

2.5.3. Spiny mouse

In recent years, the spiny mouse has emerged as a novel regenerative model among mammals, demonstrating impressive regenerative abilities across multiple organs, including the spinal cord, ear pinna, skin, kidney, and heart [34]. Following myocardial infarction, two studies showed that spiny mice exhibited preservation of heart contractile function compared to standard laboratory mice [35,36] while one study demonstrated an impressive functional improvement in spiny mice from 25 % to 30–65 % in cardiac ejection fraction [37]. If the findings in the third study hold true [37], spiny mice could be an excellent heart regeneration model. However, if the conclusions in the first two studies is true [35,36], it seems that spiny mouse hearts are more resistant to adverse remodeling, such as functional deterioration and ventricular dilation, rather than displaying robust regenerative responses, after myocardial infarction. In summary, the value of spiny mice in studying heart regeneration and repair needs to be further rigorously evaluated.

2.5.4. Rabbit

Rodent models, such as mice and rats, are widely used in cardiac disease research. However, differences in cardiac cellular and molecular characteristics between rodents and humans present challenges in translating findings to clinical settings. Rabbits serve as an intermediate model, as they share several physiological features with humans while also being closer in size to larger mammals like pigs. Due to these advantages, rabbits have been employed to study various aspects of cardiovascular diseases [38]. In rabbits, the heart exhibits regenerative capacity immediately after birth but loses this ability within the first postnatal week [39]. In P1 rabbits, post-myocardial infarction hypertrophy is not observed. Instead, they exhibit a higher number of cardiomyocyte divisions and lower fibrosis compared to P7 rabbits, highlighting their greater regenerative capacity.

2.5.5. Pig

Investigation of the neonatal cardiac regenerative window has also been extended to large mammals. Two independent laboratories induced acute myocardial infarction in neonatal pigs. They reported that thirty days following injury, heart function was reported to restore in P1 animals and partially recovered in P2 pigs, but impaired at P3 or later [40,41]. Consistently, they found little evidence of scar formation or myocardial thinning when injured at P1. Intriguingly, a follow-up study showed that apical resection in P1 pigs prolongs the cardiac regenerative window and promotes myocardial regeneration at P28 after myocardial infarction [42].

2.5.6. Sheep

Sheep, such as *Merino ewes*, are placental mammals similar to humans and may have the capacity to resist cardiac injury when they are fetuses [43]. Myocardial infarction is induced in both fetal and adolescent sheep by ligating the second diagonal of the left anterior descending artery (LAD). Three days later, they confirmed a functional reduction in the left ventricle of adolescent sheep, but not in fetal sheep possibly due to the presence of shunts. Cardiac shunts are present in the fetal stage to deliver oxygen and nutrients through bypass pathways, such as the ductus arteriosus and foramen ovale, as the lungs are not functional at that stage [44]. This shunting may be related to the lack of significant reduction in left ventricle function in fetuses after myocardial infarction. Nevertheless, animals in both groups develop significant scar tissue when examined a few months after the injury, suggesting that fetal sheep (at least at the developmental stage that was investigated) have already lost the scarless cardiac regenerative ability. In this study, cardiac shunts make it difficult to assess the level of ischemia-induced myocardial injury and the possibility of some level of cardiomyocyte regeneration. Therefore, cryoinjury, which would not be affected by cardiac shunts, could be an alternative approach to generate myocardial injury and evaluate cardiac regenerative potential in fetal and neonatal sheep.

3. Mechanisms underlying cardiac regeneration versus repair in different species

For more than twenty years, various fish, amphibian, reptile, bird, and mammalian species have been examined to assess their heart regenerative capacity. There seem to be three major mechanisms in cardiomyocyte regeneration, vasculature reformation, and communication between cardiovascular cells and other players that contribute to regenerative versus non-regenerative outcomes.

3.1. Cardiomyocyte regeneration

3.1.1. Cardiomyocyte cell cycle activity

One of the most striking differences in the cellular responses to cardiac injury in fish (zebrafish versus medaka) and frog (*Xenopus tropicalis* versus *Xenopus laevis*) studies is the level of cardiomyocyte proliferation. In both regenerative species zebrafish and *Xenopus tropicalis*, many cardiomyocytes undergo cell cycle reentry, mitosis and division to replenish the lost cardiomyocytes following myocardial injury while such an elevated cardiomyocyte cell cycle activity post injury was not observed in medaka and *Xenopus laevis* [10,20]. In medaka, it was shown that manipulations to stimulate cardiomyocyte proliferation were sufficient to rescue the phenotypes and promote heart regeneration [45], supporting that the lack of cardiomyocyte proliferation is a key contributor to defective cardiac regeneration in medaka [10].

Most mammalian cardiomyocytes withdraw from the cell cycle permanently postnatally. This event is clearly shown in mice, rabbits and pigs [4,39,42]. However, in precocial mammals such as sheep, cardiomyocytes almost completely exit the cell cycle at birth [43], but in marsupials such as opossum which are born more like embryonic placental mammals, their cardiomyocytes undergo cell cycle

withdrawal at a much later time points after birth (for opossum, it is between P32 and P56 days) [32]. Overall, the time course of cardiomyocyte cell cycle exit in different mammalian species is correlated with their limited window of regenerative capacity. Stimulation of cell cycle activity has been shown to promote adult cardiomyocyte proliferation and heart regeneration [46–49]

In mammals, the triggers of loss of cardiomyocyte proliferative potential and heart regenerative capacity have been proposed to be internal factors such as neuroendocrine activity [50,51] and external factors such as changes in oxygen concentration, fat-rich milk and metabolic shifts that occur as they transition from the maternal environment to the external world [52–54]. Indeed, neonatal mice exposed to hypoxic conditions exhibit enhanced cardiomyocyte proliferation compared to those in normal oxygen conditions [55]. Additionally, during the postnatal metabolic switch from glycolysis to fatty acid oxidation, lipid components in maternal milk have been found to play a role in cardiomyocyte maturation [53]. Interestingly, even though opossum neonates are exposed to similar environmental changes, they continue to exhibit robust cardiomyocyte proliferation and subsequent heart regenerative potential much longer after birth [32]. Moreover, most sheep cardiomyocytes exit the cell cycle before birth. Thus, the studies from these non-model organisms suggest that external factors are likely important modulators of cardiomyocyte proliferation and heart regeneration.

3.1.2. Polyploidization

In addition to cell cycle exit, most mammalian cardiomyocytes undergo polyploidization in the same time window when the heart loses its regenerative potential in opossums, mice, rabbits, pigs, and sheep [56]. Neonatal mice at postnatal day 1 (P1) have over 90 % mononucleated cardiomyocytes. However, within a week, approximately half of the cardiomyocytes become binucleated, and eventually, over 90 % will be binucleated, with less than 10 % remaining mononucleated [57]. A similar trend is observed in pigs [58], while in human cardiomyocytes, the trend is present but not as pronounced [59]. Polyploidization could occur as a result of failure in cytokinesis (multinucleated), karyokinesis and both. The exact purpose of cardiomyocyte polyploidization is not known, but it has been proposed that cardiomyocyte polyploidization may enhance contractile strength, increasing the heart's pumping capacity and providing greater resistance to cellular damage [60]. The additional genetic material in polyploid cells may act as a buffer against genomic DNA damage [61]. This could explain why cardiomyocytes exit the cell cycle, as they attain a higher level of protection compared to other cell types, even though their inability to generate new cells following severe damage is a trade-off.

The function of cardiomyocyte polyploidization to impede heart regeneration and its divergence in different vertebrate species have been shown in some recent studies. First, zebrafish have almost exclusively diploid cardiomyocytes in the adult heart. Genetic manipulations to block cytokinesis yielded zebrafish mutants with polyploid cardiomyocytes that were unable to complete heart regeneration [62]. Secondly, a comprehensive survey of 120 inbred mouse strains identified variations of cardiomyocyte ploidy in these strains [63]. Mouse strains with higher percentages of diploid cardiomyocytes exhibit better cardiac regenerative potential [63]. Thirdly, studies of cardiomyocyte ploidy across 41 vertebrate species revealed that certain monotreme, edentate, cetacean, and chiropteran species retain abundant diploid cardiomyocytes [50]. Moreover, the frequency of polyploid cardiomyocytes is inversely correlated with standard metabolic rates, body temperature and circulating hormones [50]. Inhibition of two major thermogenic pathways in postnatal mice led to lower body temperatures, a 12-fold increase of diploid cardiomyocytes and extension of cardiac regenerative window [51]. These findings suggest that cardiomyocyte polyploidization and loss of regenerative capacity may be a trade-off of endothermy acquisition in both development and evolution.

3.2. Vascular regeneration

The heart's ability to regenerate after injury is a central topic in regenerative biology and medicine, particularly in the context of ischemic heart diseases such as myocardial infarction. One critical aspect of cardiac regeneration is vascularization [64], the process by which new blood vessels form, ensuring the delivery of oxygen and nutrients to damaged tissues. Vascularization not only promotes tissue survival but also plays an essential role in the regenerative response across species [8] (Fig. 2). However, this capacity for vascularization and subsequent heart regeneration varies significantly among species and is often lost in mammals after early developmental stages.

Among vertebrates, zebrafish (*Danio rerio*) are well-known for their extraordinary regenerative capacity. Zebrafish can fully regenerate their hearts following injury through a process that involves the proliferation of both cardiomyocytes and endothelial cells, leading to the formation of new blood vessels [65]. New coronary vasculature formation depends on Cxcr4a-Cxcl12b [66], Vegfr-VEGF [67] and other signaling pathways as zebrafish with mutation in these components showed minimal or abnormal vascularization and inhibition of heart regeneration. These findings highlight the critical role of new vasculature formation in cardiac regeneration. As widely used models for studying heart regeneration, mice exhibit a transient regenerative ability early in life [4]. During the first postnatal week, mouse hearts are capable of limited regeneration, particularly through the formation of collateral arteries [64]. Collateral arteries act as natural bypasses that redirect blood flow to ischemic areas after myocardial infarction, which is crucial for tissue survival during regeneration. However, this ability to form new blood vessels is lost shortly after birth, around postnatal day 7 (P7), marking the end of the mouse heart's regenerative window [64]. Strikingly, exogenous administration of Cxcl12 promotes collateral artery formation and facilitates cardiac repair [64,68], even though *Cxcl12* expression itself is induced by hypoxic conditions due to ischemic injury in the postnatal stage, which cannot regenerate hearts. Thus, there may be other mechanisms that can enhance collateral artery formation, ultimately promoting heart regeneration. Guinea pigs share physiological similarities with humans in many organs [69] and are used for translational purposes in medicine [70]. Most mammals, such as rats, rabbits, and others, develop infarction immediately after ischemic injury, including arterial occlusion. Interestingly, however, guinea pigs are not affected by the occlusion [71]. This is because guinea pigs have well-developed collateral arteries, which can deliver oxygen and nutrients from other areas to the infarcted region. These observations

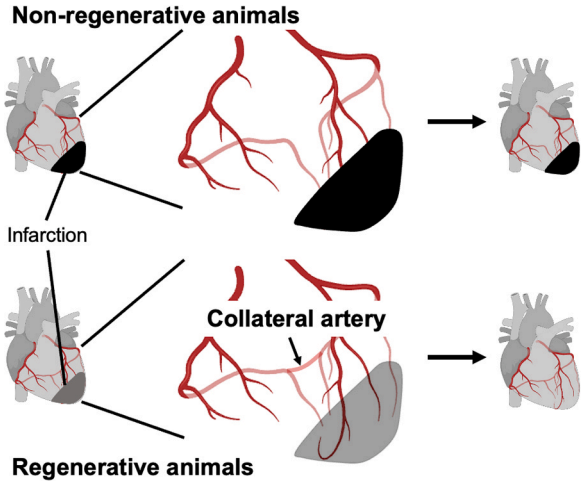


Fig. 2. Collateral artery formation is a key event for heart regeneration. In non-regenerative animals or individuals, there is either an absence of innate collateral arteries or an inability to develop these arteries after injury. However, the mechanisms behind collateral artery formation are largely unknown.

strongly suggest that innate collateral arteries have a significant protective effect against heart injury, and that promoting neovascularization may also be beneficial in healing injured hearts. Humans, like most mammals, have a very limited ability to regenerate cardiac tissue after injury. Myocardial infarction often results in extensive tissue damage, scar formation, and progressive heart failure, primarily due to the inability to efficiently revascularize the injured myocardium. Interestingly, humans, particularly those with coronary artery disease, develop collateral arteries as the heart adapts to the ischemic environment, forming these vessels to support the ischemic myocardium [72]. Although most cases of acute infarction are treated with catheterization surgery [73], there are instances where this procedure cannot be applied. For these patients, several established methods exist to promote the formation of collateral arteries [74]. In summary, the mechanisms underlying revascularization remain largely unknown and are less studied compared to those regulating cardiomyocyte status. Since promoting vascularization is a critical strategy for repairing injured hearts, elucidating these mechanisms is essential for advancing therapeutic approaches.

3.3. Communication between cardiovascular cells and other cellular players

3.3.1. Inflammatory cells - cardiomyocytes

Medaka fail to increase cardiomyocyte proliferation following injury, unlike zebrafish which demonstrate significant elevation in proliferation [45]. To understand the upstream differences, research utilizing comparative transcriptomic analysis between zebrafish and medaka after cryoinjury has provided insights into the underlying mechanisms of their differing regenerative abilities, especially focusing on immune response [45,75]. Following heart injury, immune cells such as neutrophils and macrophages are rapidly activated and recruited to the injured area. In zebrafish hearts, neutrophils are cleared out more quickly, allowing macrophages to persist in the injured area for a longer duration compared to medaka hearts. This timing of macrophage recruitment plays a crucial role in the process of heart regeneration. The accumulation of macrophages is delayed in medaka. Stimulating Toll-like receptor signaling by poly I:C in medaka leads to enhanced cardiomyocyte proliferation, neovascularization, neutrophil clearance, and scarring resolution [45]. In addition to immune responses, several signaling pathways related to hormonal and metabolic regulation, such as steroid hormone signaling [76] and retinoid signaling—characterized by the local expression of *Raldh2* in endocardial tissue [19,77]—exhibit differential changes between zebrafish and medaka. Interestingly, despite sharing a low metabolic state, hypoxic conditions, and similar size and morphology, their regenerative capacities are entirely different. There remains much to explore regarding the mechanisms underlying this disparity, and future studies will help elucidate the precise factors that could enhance heart regeneration in species with limited regenerative potential.

3.3.2. Cardiomyocytes - non-cardiomyocytes

Although cavefish and surface fish show a comparable number of proliferating cardiomyocytes following cardiac injury, cavefish demonstrate more persistent or even higher levels of proliferation in non-cardiomyocyte cells than surface fish. This paradox suggests that simply having proliferating cells does not equate to effective heart regeneration in cave fish. A genetic comparison between surface fish and cavefish has revealed key insights into why cavefish lose their regenerative capacity. Specifically, cavefish exhibit higher expression levels of genes related to inflammation and fibrosis, coupled with lower expression of metabolism-related genes. Furthermore, the study identified a gene called Irrc10 as a regeneration suppressor through transcriptome comparisons between surface and cave populations [14]. Since surface fish and cavefish belong to the same species, the researchers crossed them to create an F2 generation and identified three genetic loci that are important for heart regeneration [14]. Although this species takes about a year to reach adulthood, this comparative technique represents a powerful tool for researchers. It effectively circumvents challenges associated with poor genome annotation and the differing gene sets found when comparing regenerative and non-regenerative animals. By utilizing this approach, scientists can gain a deeper understanding of the genetic basis of heart regeneration. In the future, this work may reveal how non-cardiomyocytes either downstream of or in parallel with cycling cardiomyocytes account for the defective heart regeneration in cavefish.

4. Concluding remarks

The ability of organisms to regenerate heart tissue is closely tied to the status of cardiomyocytes and vasculature, with significant differences observed across species. Mice and humans lose their ability to activate cardiac proliferation and form collateral vessels early in life, whereas non-mammalian species such as zebrafish and salamanders retain this ability throughout their lifespan. Although cardiomyocyte proliferation is often considered an indicator of regenerative capacity, findings from cavefish suggest that this is not always the case. Studying the molecular pathways that regulate cardiomyocyte status, including proliferation, and vascularization in these diverse models could provide critical insights into potential therapeutic targets for treating heart disease in humans. Strategies aimed at enhancing cardiomyocyte proliferation, promoting collateral vessel formation, and reactivating regenerative pathways in the human heart hold great promise for the future of cardiovascular medicine.

Declaration of Competing Interest

None

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